M7 Results Summary

Clean accuracy

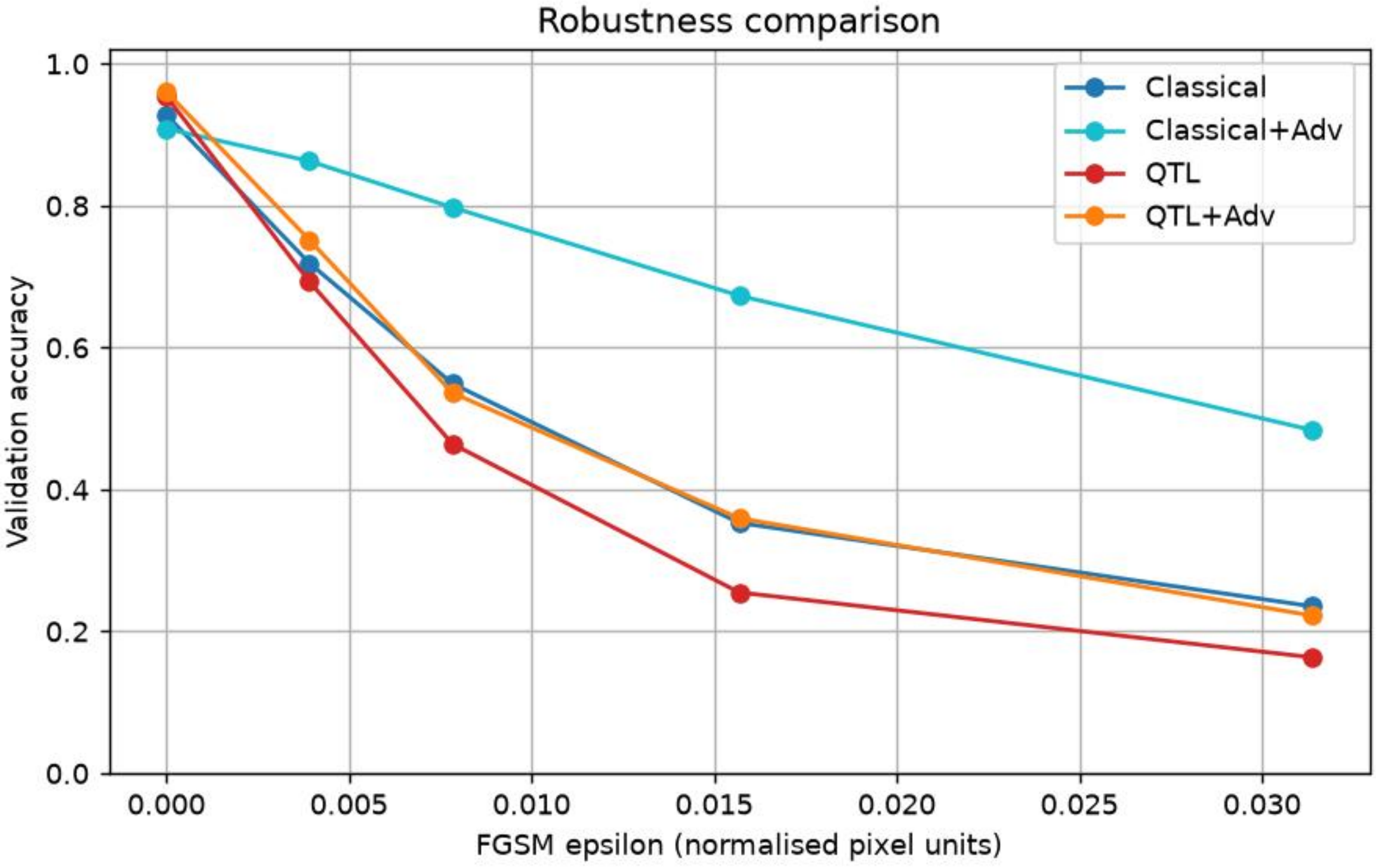
Model	Clean acc	F1
Classical	0.9281	0.9278
Classical+Adv	0.9085	0.9083
QTL	0.9542	0.9541
QTL+Adv	0.9608	0.9605

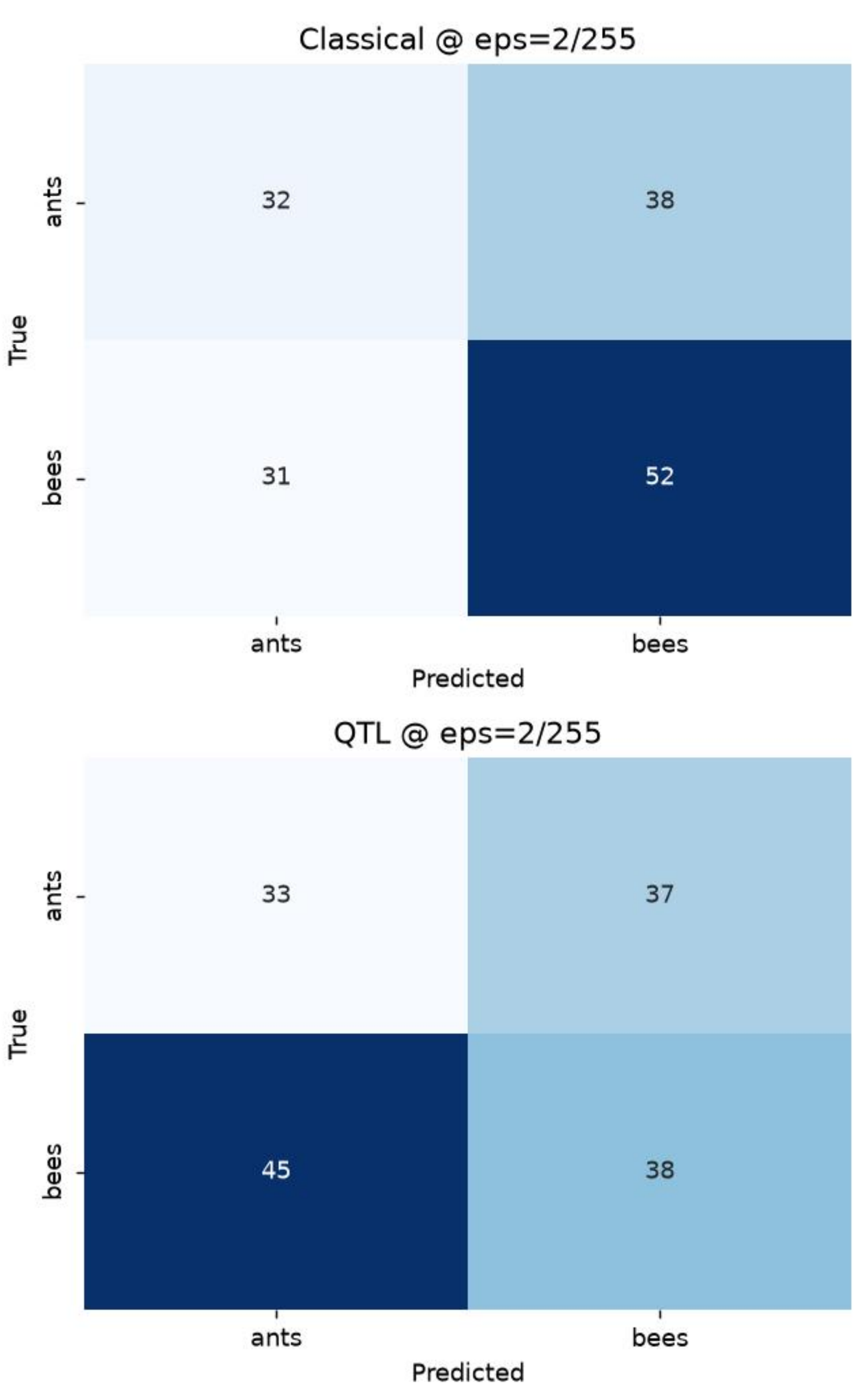
Robustness (accuracy under FGSM)

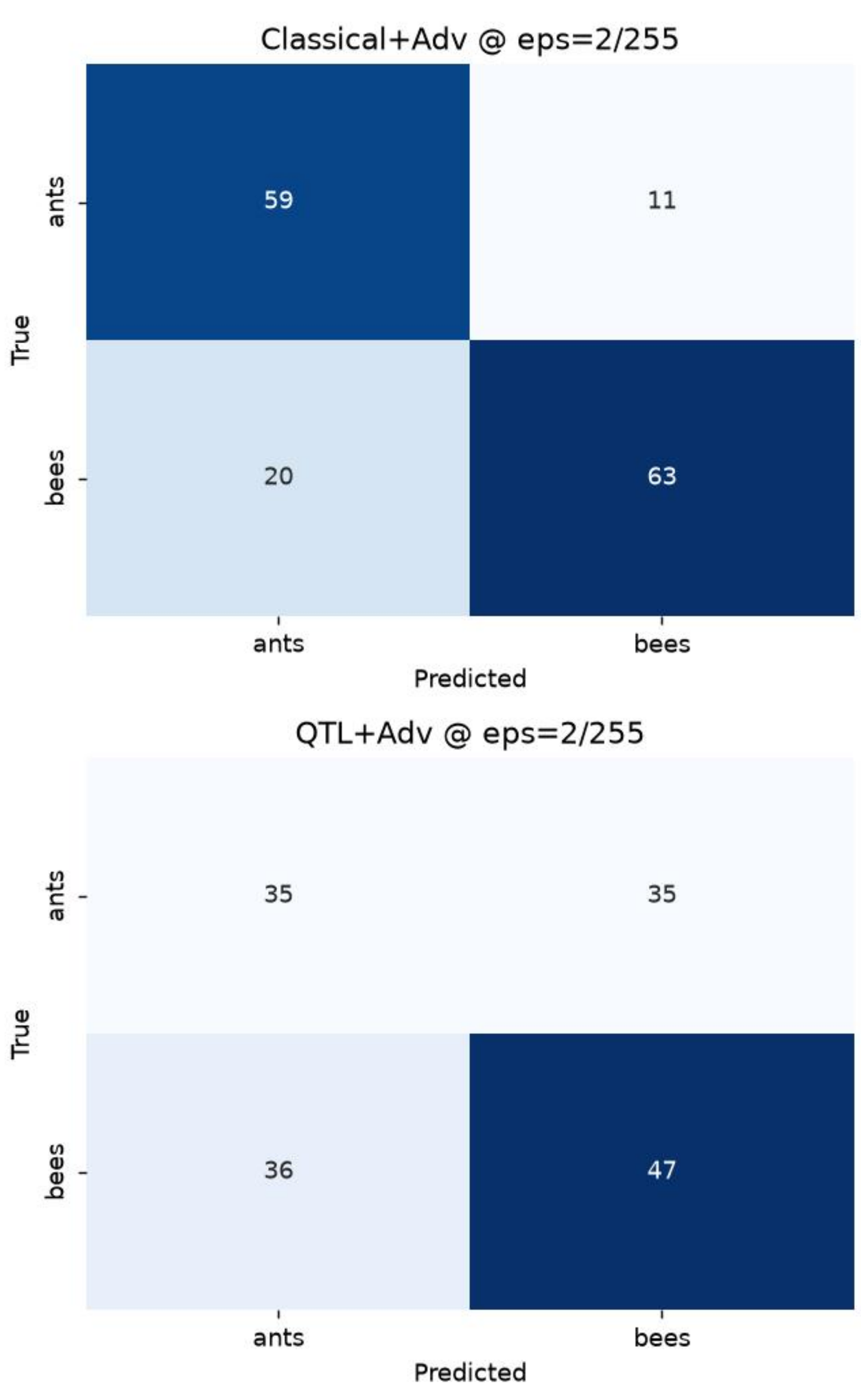
Model	eps=0.0000	eps=0.0039	eps=0.0078	eps=0.0157	eps=0.0314
Classical	0.9281	0.7190	0.5490	0.3529	0.2353
Classical+Adv	0.9085	0.8627	0.7974	0.6732	0.4837
QTL	0.9542	0.6928	0.4641	0.2549	0.1634
QTL+Adv	0.9608	0.7516	0.5359	0.3595	0.2222

Attack Success Rate

Model	eps=0.0000	eps=0.0039	eps=0.0078	eps=0.0157	eps=0.0314
Classical	0.0000	0.2254	0.4085	0.6197	0.7465
Classical+Adv	0.0000	0.0504	0.1223	0.2590	0.4676
QTL	0.0000	0.2740	0.5137	0.7329	0.8288
QTL+Adv	0.0000	0.2177	0.4422	0.6259	0.7687


Robustness comparison
Classical
Classical+Adv
QTL
QTL+Adv
Validation accuracy
1.0
0.8
0.6
0.4
0.2
0.0
0.000
0.005
0.010
0.015
0.020
0.025
0.030
FGSM epsilon (normalised pixel units)


Classical @ eps=2/255
ants
bees
True
32
38
31
52
ants
bees
Predicted


Classical+Adv @ eps=2/255
ants
bees
True
59
11
20
63
ants
bees
Predicted

QTL @ eps=2/255
ants
bees
True
33
37
45
38
ants
bees
Predicted

QTL+Adv @ eps=2/255
ants
bees
True
35
35
36
47
ants
bees
Predicted

Clean (top) vs FGSM-adversarial (bottom) images — eps=4/255

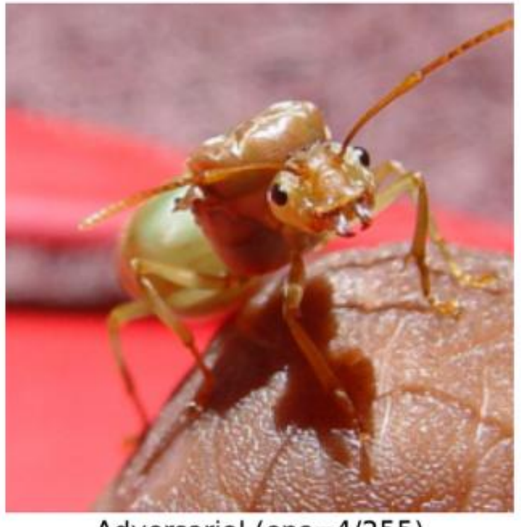
Clean | true = ants
Classical: ants
Classical+Adv: ants
QTL: ants
QTL+Adv: ants



Clean | true = ants
Classical: ants
Classical+Adv: ants
QTL: ants
QTL+Adv: ants



Clean | true = ants
Classical: ants
Classical+Adv: ants
QTL: ants
QTL+Adv: ants



Clean | true = ants
Classical: ants
Classical+Adv: ants
QTL: ants
QTL+Adv: ants



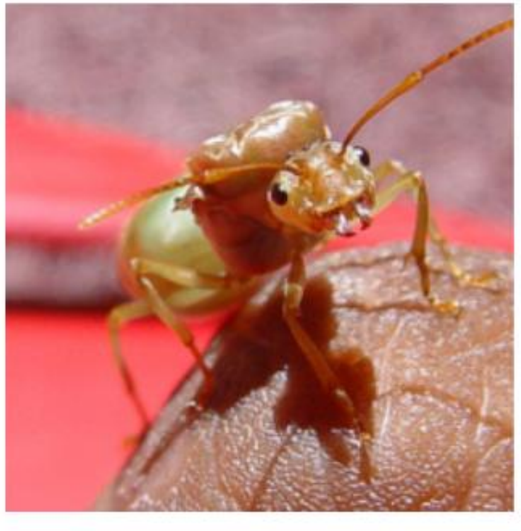
Adversarial (eps=4/255)



Adversarial (eps=4/255)

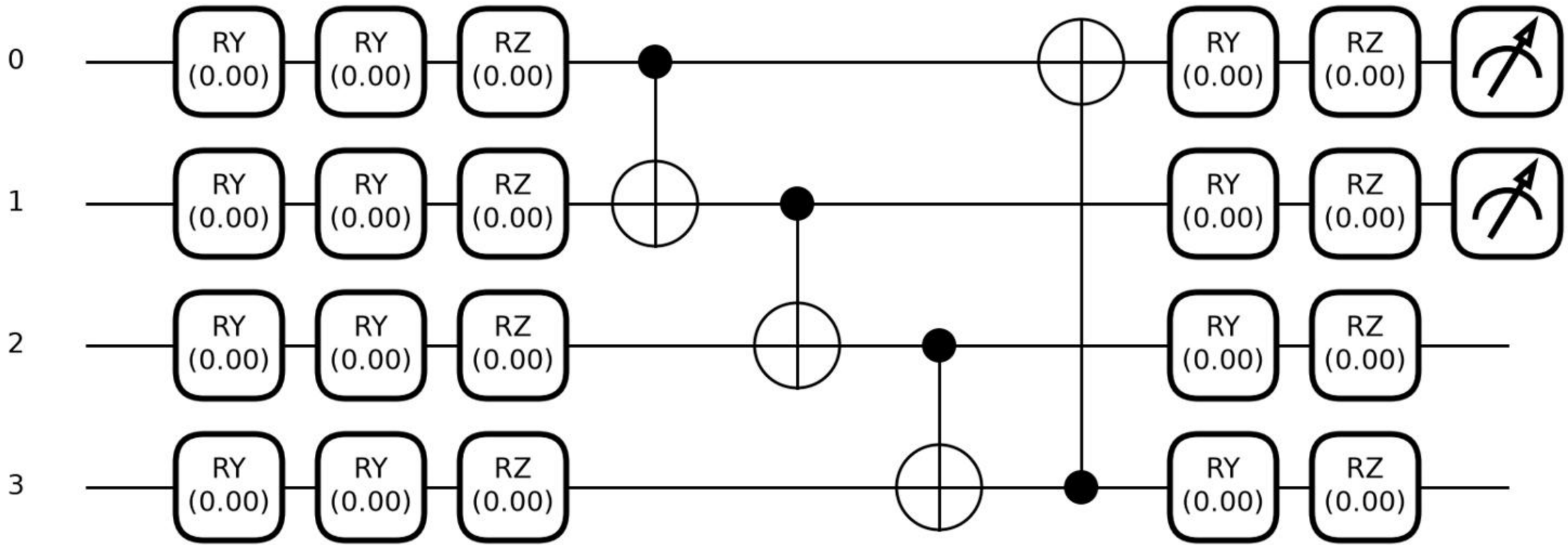


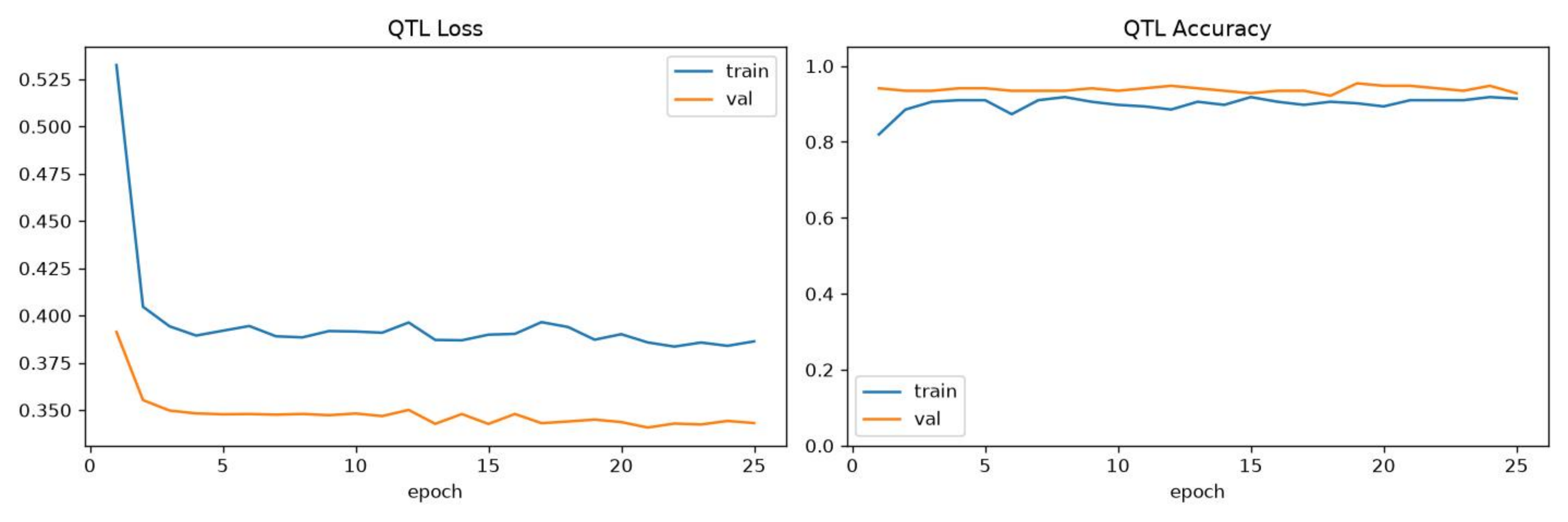
Adversarial (eps=4/255)

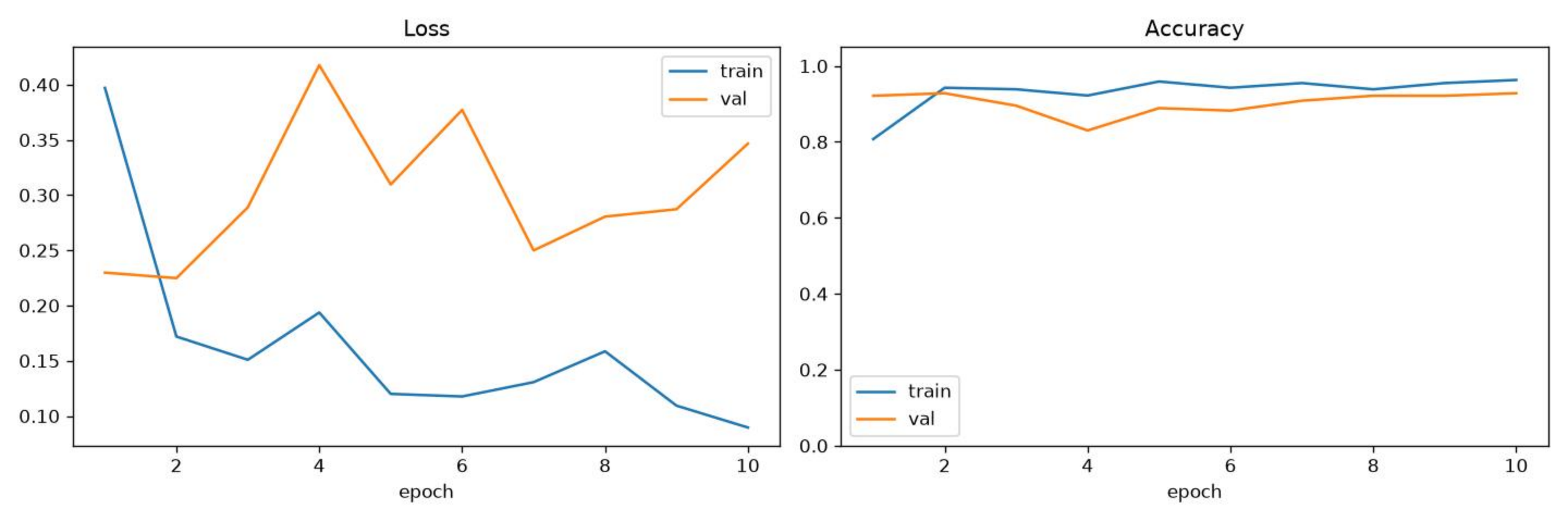


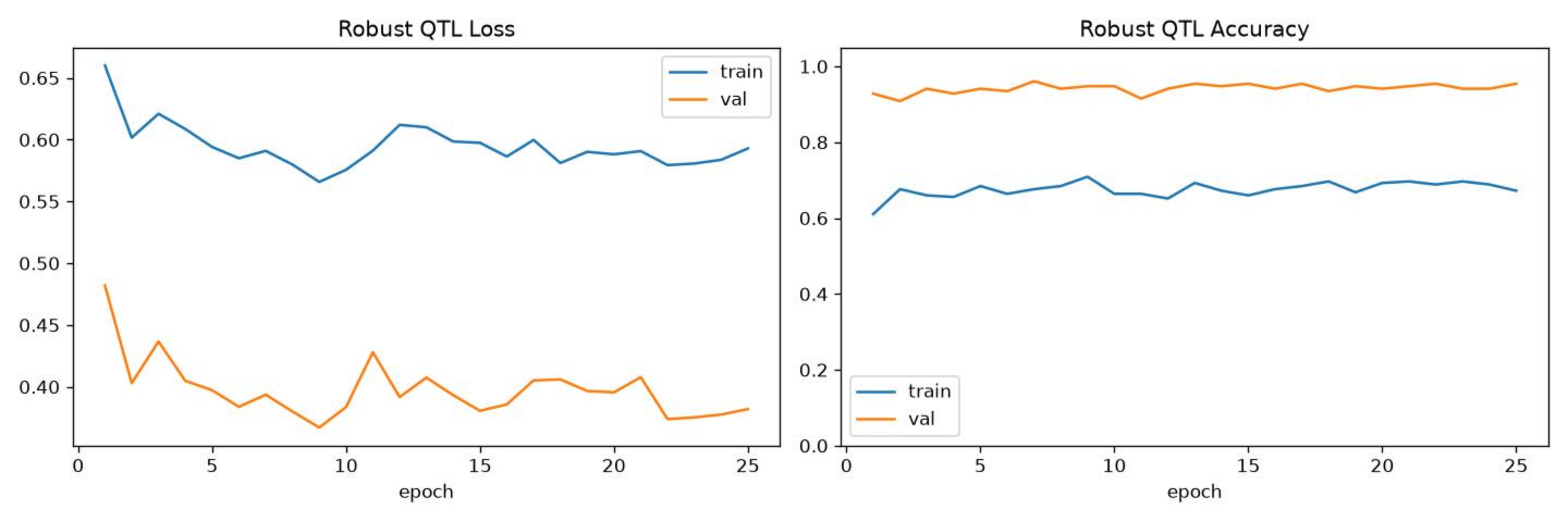
Adversarial (eps=4/255)




0
1
2
3
RY
(0.00)
RZ
(0.00)


QTL Loss
train
val
0.525
0.500
0.475
0.450
0.425
0.400
0.375
0.350
0
5
10
15
20
25
epoch
QTL Accuracy
train
val
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25
epoch


Loss
train
val
0.40
0.35
0.30
0.25
0.20
0.15
0.10
2
4
6
8
10
epoch
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
train
val
2
4
6
8
10
epoch


Robust QTL Loss
train
val
0.65
0.60
0.55
0.50
0.45
0.40
0
5
10
15
20
25
epoch
Robust QTL Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
train
val
0
5
10
15
20
25
epoch